

GMRS-TTY

User Manual — Full Reference

A TTY-style accessibility communicator for the General Mobile Radio Service (GMRS) and Family Radio Service (FRS), built for deaf, hard-of-hearing, and mute operators.

Live transcription · offline TTS · FCC Part 95 ID rules built in

Read the keyboard-shortcut reference on page 4 if you want to skip the tour and start operating immediately.

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1. About this manual

This manual is the full operating reference for GMRS-TTY, a cross-platform desktop application that lets deaf, hard-of-hearing, and mute operators participate in two-way voice radio conversations over GMRS and FRS. The program is written in Python with PySide6, runs fully offline once installed, and ships every feature behind the FCC Part 95 rule set so operators can stay legal without memorizing the regulations.

The manual is organized as a reference rather than a tutorial. Section 5 walks every region of the main window in left-to-right, top-to-bottom order. Sections 6–9 document each dialog. Section 10 is the keyboard cheat sheet. Sections 11–16 cover the deeper behaviors (service mode, PTT modes, the audio pipelines, callsign verification, and FCC compliance). Section 19 is troubleshooting. Sections 20–21 give the on-disk file formats and a glossary.

If you only have five minutes: install the dependencies (section 3), copy `config.example.json` to `config.json` and edit your callsign / name / location, drop one Piper voice into `Voices/`, run `python bootstrap_models.py`, then launch with `python main.py`. Press **Alt+L** to listen and type into the message box to transmit.

2. System requirements

Operating system

- **Linux** — Debian 12+, Ubuntu 22.04+, Raspberry Pi OS (Bookworm). PortAudio dev libs required: `sudo apt install libportaudio2 portaudio19-dev`. On PipeWire systems also install `pulseaudio-utils` so the `parec` capture path is available — the app prefers it because PortAudio's PipeWire-via-ALSA bridge can silently deliver flat-zero audio on PipeWire 1.4.
- **Windows** — Windows 10 or 11. PortAudio ships in the wheel; no extra system libraries needed.
- **macOS** — Not formally targeted, but Python 3.11+ with PortAudio available through Homebrew usually works.

Hardware

- **Microphone** — Anything Linux/Windows recognizes — a USB headset, the built-in laptop mic, or a sound-card channel cabled from your radio's speaker output.
- **Speaker / radio audio path** — A speaker for live listening, or a USB sound card / Signalink / Digirig for feeding TTS audio directly into your radio.
- **Optional: PTT cable** — A USB-serial adapter (FTDI is the most common) for keying PTT via RTS/DTR. Not needed if your radio has VOX or you key manually.

Software

- Python 3.11 or newer (3.13 recommended).
- ~1 GB free disk for Python dependencies (CTranslate2, ONNX Runtime, PySide6) plus 75–150 MB for the Whisper STT model.
- Internet access **once** to install pip packages and bootstrap the STT model. After that the core radio workflow needs no network at all.

3. Installation

Five steps from a fresh clone to a working radio session.

3.1 Clone and create a virtual environment

```
git clone <repo-url> GMRS-TTY
cd GMRS-TTY

python3 -m venv .venv
source .venv/bin/activate          # Linux / macOS
# .venv\Scripts\activate          # Windows

pip install -r requirements.txt
```

3.2 Drop in a Piper voice

Download at least one Piper ONNX voice and its matching .json config file into a Voices/ directory at the project root. The Configuration dialog reads this folder at launch and lists every voice it finds.

```
Voices/
■■■ en_US-ryan-high.onnx
■■■ en_US-ryan-high.onnx.json
■■■ en_US-amy-medium.onnx
■■■ en_US-amy-medium.onnx.json
```

Voices: <https://github.com/rhasspy/piper/blob/master/VOICES.md>. Most voices are MIT-licensed; en_US-libritts-high is CC BY 4.0 and requires attribution if you redistribute it.

3.3 Bootstrap the speech-to-text model

The Whisper STT model is not bundled in the repo. Fetch it once on an internet-connected machine:

```
python bootstrap_models.py          # default: small.en
python bootstrap_models.py --model base.en    # smaller, faster, less accurate
python bootstrap_models.py --model medium.en  # higher accuracy, slower
```

This populates Models/STT/<model_name>/ with the faster-whisper CTranslate2 artifacts. The app loads from there on Listen and never attempts network access — if the directory is missing, listening fails fast with an instruction to run the bootstrap.

For air-gapped installs: run the bootstrap once on an internet-connected machine, then copy the entire Models/ directory (alongside the source) to the offline target. Silero VAD and Piper voices ship as local files already, so no other fetches are involved.

3.4 Configure

```
cp config.example.json config.json
$EDITOR config.json    # set your callsign, name, location, voice
```

You can also leave the file at its defaults and edit everything through the in-app Configuration dialog after first launch.

3.5 Run

```
source .venv/bin/activate  
python main.py
```

4. First run & configuration

On first launch the app reads `config.json` from the working directory. If your callsign is still `YOUR_CALL` the header will display that literal string — open Settings → Configuration (or click the gear icon in the top-right of the service row) and fill in:

- **Callsign** — Your FCC GMRS callsign (e.g. WSLZ233). Saved uppercased.
- **Name** — Your operator name. Appears in the callsign preface and the station-ID button output.
- **Location** — Free-form city / state. Used by the standalone **This is** ID announcement.
- **Voice Model** — Pick the Piper voice you dropped into Voices/. Click **Test** to hear a sample before committing.
- **Speech Rate** — Slider from 0.70× (faster) to 1.50× (slower); 1.00× is the voice's native pace.
- **Input Device** — Microphone the Listen button captures from. *System Default* works for most setups.
- **Output Device** — Where TTS audio is played — choose a USB sound card / Signalink / Digirig channel here to feed your radio directly.
- **VAD Threshold** — Silero VAD sensitivity (0.10–0.95). Lower is more sensitive (catches quiet signals but more false starts); higher is stricter.
- **Time Format** — 24-hour or 12-hour AM/PM timestamps on RX lines.
- **Filter profanity** — On by default. Masks the f/s-words and similar with asterisks in both RX transcripts and outgoing TX before TTS speaks them.
- **PTT Mode** — Manual, VOX, or USB FTDI / Serial. See section 12.

Click **OK**. The header updates, the dropdown lists the new voice, and the listener restarts automatically if you changed the input device or VAD threshold.

Configuration is stored as plain JSON in `config.json` next to `main.py`. You can hand-edit it between runs; the in-app dialog and the file are interchangeable.

5. Main window tour

The main window is laid out top-to-bottom in five horizontal bands: the **service toggle row**, the **header strip**, the **chat area** with its toolbar, the **pending stations and quick-messages strip**, and the **transmit row** with the **This is** button beneath it. The menubar holds Settings, and the status bar carries the live online/offline indicator. Minimum window size is 720×520 to guarantee no clipping at high-DPI / large-font settings; the default is 800×600.

5.1 Service toggle row (top)

- **Service:** — Label.
- **GMRS / FRS** — Segmented radio buttons (Alt+G / Alt+F). Choose your operating service. The selection persists to `radio_service` in `config.json`. See section 11 for the full behavior diff.
- **Q icon (right cluster)** — Bold capital Q. Opens the Quick Messages editor — same destination as Settings → Quick Messages.
- **Person icon** — Bust-in-silhouette glyph. Opens Contacts — same destination as Settings → Contacts or Ctrl+B. Disabled in FRS mode (no callsigns).
- **Gear icon** — Cog wheel. Opens Configuration — same destination as Settings → Configuration or Ctrl+,. Enabled in both modes.

5.2 Header strip

Bold panel just under the service row. In GMRS mode it reads `Station: WSLZ233 | Operator: Benjamin | Location: Lansing, MI`. In FRS mode the station segment is replaced with FRS Mode because FRS has no callsign requirement.

5.3 Chat area

- **Clear chat button** — Right-aligned, sits above the log. Alt+C on the button, or Ctrl+K from anywhere. Asks for Yes/No confirmation then wipes the log. Chat history is in-memory only — it cannot be recovered once cleared.
- **Conversation log** — Timestamped messages. Incoming lines are green `[RX HH:MM:SS] :`. Outgoing lines are blue `[TX to <target>] :`. Errors say `Error:` or `Failed:` in the text — state is never carried by color alone.
- **Auto-tail** — New messages keep the viewport pinned to the bottom while you're caught up. Scroll up to re-read older context and incoming traffic will *not* yank you back.
- **Callsign pills** — Any callsign that matches a saved contact is rendered with an amber, bold pill background. Hover for every operator linked to that callsign — useful for family-shared GMRS calls. A green ✓ appears immediately after the callsign when the contact has a confirmed FCC license match (hover the check for the FCC license verified tooltip).

5.4 Transmit row

- **Listen button** — Toggle button (Alt+L or Ctrl+L). Starts/stops microphone capture and live transcription. Loads the Whisper model from `Models/STT/<model>/` on first start; subsequent toggles are instant.

- **Input level meter** — Thin progress bar immediately to the right of Listen. Real-time peak amplitude of the captured audio. Use it to verify your mic / cable / device is wired up — if it stays at zero while you key audio into the radio, the app isn't getting audio. Stays at zero when Listen is off.
- **Target dropdown** — Pick a contact callsign or *All*. Entries are sorted alphabetically; *All* is pinned at the top. Family-shared GMRS calls appear on separate rows like WSLZ233 (Eliza) / WSLZ233 (Jennifer), and the preface speaks the exact operator name from the row you selected.
- **Message box** — Text input. Type your message and press Enter or click Transmit. Placeholder reads *Type your message here....*
- **Transmit button** — Alt+T or Ctrl+Return / Ctrl+Enter. Sends the message through the TX pipeline: shorthand expansion, profanity masking, callsign framing, 15-minute ID check, PTT keying, TTS synthesis, and STT auto-pause until the unkey.

5.5 Standalone “This is” button

Sits in its own row beneath Transmit. Alt+I or Ctrl+I. Sends a standalone station ID — *This is <CALL>, <NATO phonetic>. <name> from <location>.* — without needing to type anything. Resets the 15-minute ID timer. Disabled in FRS mode (FRS has no ID rule); hover the disabled button for the explanation.

5.6 Pending stations bar

Sits between the chat log and the quick-messages strip. Hidden when no pills are pending. Yellow pill buttons appear when a new GMRS callsign is detected on RX:

- **Click** a pill to open the Add Station dialog (section 9) pre-filled with the detected name and location.
- **Right-click** or long-press a pill to dismiss that one without adding it.
- **Dismiss all** (Alt+D) appears on the right edge whenever any pill is present — clears every pending pill at once.
- Pills wrap to additional rows up to a maximum of three; past that a vertical scrollbar appears so the chat area doesn't get squeezed.

5.7 Quick-messages strip

Strip of one-click buttons between the pending bar and the transmit row. Each button rides the standard TX pipeline. Curly-brace tokens like {N} in QSY to channel {N} prompt for a value before transmitting. The first nine buttons are bound to **Alt+1** through **Alt+9**. Edit the list from Settings → Quick Messages or the Q icon.

The seed list is: *Radio check, Loud and clear, Standing by, Acknowledged, Say again, QSY to channel {N}, Clear, Monitoring, Net check-in, Emergency traffic.*

5.8 Status bar

Carries a permanent **Online / Offline** indicator on the right edge. Green ● for online, amber ■ for offline. Updates every 30 seconds. The indicator is the user-visible side of the contract for opt-in network features — when offline, the FCC verification button disables and save-time verification is skipped. Hidden in FRS mode.

5.9 Menubar

Single **Settings** menu (Alt+S). Contains:

- **Configuration... (Alt+C or Ctrl+,)** — Opens the Configuration dialog (section 6).
- **Contacts... (Alt+N or Ctrl+B)** — Opens the Contacts dialog (section 7). Disabled in FRS mode.
- **Quick Messages... (Alt+Q)** — Opens the Quick Messages editor (section 8).
- **Clear chat (Alt+R or Ctrl+K)** — Erases every message from the log after a Yes/No confirmation.

6. Configuration dialog

Opened from Settings → Configuration, the gear icon, or Ctrl+,. Minimum width 420 px. All fields are mnemonic-linked (Alt+letter focuses the underlined field). The OK button is disabled until the background device-enumeration thread finishes; while loading you'll see *Loading devices...* in the device dropdowns.

Field	Type	Behavior
Callsign (Alt+C)	Text	Your FCC GMRS callsign. Saved uppercased and stripped.
Name (Alt+N)	Text	Operator name. Used in callsign preface and ID.
Location (Alt+L)	Text	City, state. Used by the standalone This is announcement.
Voice Model (Alt+V)	Dropdown + Test	Lists every .onnx file in Voices/. Test button (Alt+T) synthesizes a short sample at the current speech rate, played on the currently selected output device.
Speech Rate (Alt+R)	Slider 0.70×–1.50×	Maps to Piper length_scale. 1.00× normal, lower = faster, higher = slower. Step 0.05. Stored as tts_length_scale.
Input Device (Alt+I)	Dropdown	Microphone for capture. System Default plus every device PortAudio reports. Changing this restarts the listener automatically.
Output Device (Alt+O)	Dropdown	Where TTS audio plays. Pick a USB sound card to feed your radio directly. System Default uses the OS default sink.
VAD Threshold (Alt+D)	Spin 0.10–0.95 step 0.05	Silero VAD speech probability cutoff. Lower = more sensitive, higher = stricter. Default 0.50. Changing this restarts the listener.
Time Format (Alt+F)	Dropdown	24-hour (14:32:15) or 12-hour (2:32:15 PM) for RX timestamps.
Filter profanity (Alt+Y)	Checkbox	Mask strong language with asterisks. Default on. Applies to both RX transcripts and TX messages before TTS speaks them.
PTT Mode (Alt+P)	Dropdown	Manual / VOX / USB FTDI / Serial. See section 12.
Serial Port (Alt+S)	Text	Only enabled in USB FTDI mode. e.g. /dev/ttyUSB0 or COM3.
Control Line (Alt+E)	Dropdown	Only enabled in USB FTDI mode. RTS or DTR.

On save the dialog returns a sanitized config object: callsign uppercased, name/location/serial-port stripped of leading and trailing whitespace, length scale rounded to two decimals.

7. Contacts dialog

Opened from Settings → Contacts, the person icon, or Ctrl+B. Minimum size 820×360. Six columns of editable data; **Verified** is read-only and reflects the FCC verification gate.

Columns

- **Callsign** — Saved uppercased. Required — rows with empty callsign are dropped on save.
- **Name** — Free-form operator name. Used by the callsign-match side of FCC verification (“Tim” matches “Surname, Timothy L” in the license record).
- **Location** — Free-form city / state. Auto-backfilled from the FCC city on a successful verification when the row had no location of its own; values you typed are never overwritten.
- **GMRS** — Cross-reference callsign — the operator’s GMRS call if the primary callsign is their HAM call. Auto-populated from FCC related records (service code ZA) on verification. Hand-editable. Uppercased on save.
- **HAM** — Cross-reference Amateur callsign. Auto-populated from FCC related records (service codes HA / HV). Hand-editable. Uppercased on save.
- **Verified** — Read-only. Green ✓ when the callsign is in the active FCC database AND the contact’s name matches a token in the licensee’s name. Hover for the licensee, GMRS / HAM cross-refs, and the timestamp of the last successful lookup.

Buttons

- **Add Contact** — Append a row with default values NEW_CALL / New Name. Edit in place.
- **Remove Selected** — Remove the currently selected row.
- **Sort by Suffix (Alt+S)** — View-only reorder by the last 3 digits of each callsign. Useful for spotting consecutive licenses or family clusters. *ALL* stays at the top. Clicking OK still saves the list alphabetically.
- **Verify all (Alt+V)** — Check every not-yet-verified row against the FCC database. Already-verified rows whose callsign and name match what was loaded are cached and skipped. Disabled when the app is offline; hover the disabled button for the reason.
- **OK / Cancel** — Standard. OK runs the same verification gate as Verify all (newly added rows, edits, and previously-failed lookups all get a fresh round trip; cached rows are skipped) and saves to `contacts.json`.

Verification semantics

- A row earns a green ✓ when (a) the callsign is in the active FCC database AND (b) the contact’s name matches a token in the licensee’s name. Family members on a shared GMRS callsign whose name doesn’t match the licensee remain unverified.
- GMRS / HAM cross-references are *only* written when the name matches the licensee. A family-member row keeps its own GMRS / HAM fields untouched because those callsigns describe the licensee, not the family member.

- Offline behavior: Verify all disables, save-time verification is skipped, and previously-earned green checks are preserved untouched — a transient outage will never wipe verified state.

8. Quick Messages dialog

Opened from Settings → Quick Messages or the Q icon in the top-right of the service row. Minimum size 520×360. Single-column editable table of phrases plus four management buttons.

- **Add (Alt+A)** — Append a blank row and immediately enter edit mode on the new cell.
- **Remove Selected (Alt+R)** — Delete the currently selected row.
- **Move Up (Alt+U) / Move Down (Alt+D)** — Reorder the selected row. Order matters: the first nine rows get **Alt+1...Alt+9** shortcuts; everything past slot nine is mouse-only.
- **OK / Cancel** — OK saves the trimmed phrase list to `config.json` under `quick_messages`. Blank rows are dropped so the strip never shows an unlabeled button.

Placeholder tokens

Wrap a token in curly braces to prompt for a value at TX time. `QSY to channel {N}` opens a small input dialog asking for *N*; the substituted phrase then rides the standard TX pipeline. Multiple tokens are prompted in order. Cancelling any prompt aborts the transmission.

9. Add Station dialog

Opens when you click a yellow pill on the pending-stations bar. Compact three-field form with three QLineEdit inputs:

- **Callsign (Alt+C)** — Pre-filled with the detected callsign in canonical (compact, uppercase) form.
- **Name (Alt+N)** — Pre-filled if the heuristic detected a name in the surrounding transcription.
- **Location (Alt+L)** — Pre-filled if a location was detected.

Click **OK** to append the new contact to `contacts.json`; the target dropdown is rebuilt and any historical RX lines mentioning that callsign retroactively gain the amber-pill highlight. The originating pill is removed automatically. **Cancel** leaves the pill in place so you can come back to it later.

10. Keyboard shortcuts & mnemonics

Global shortcuts

Action	Shortcut
Toggle Listen	Ctrl+L
Transmit message	Ctrl+Return / Ctrl+Enter
Send standalone station ID	Ctrl+I
Clear chat (with confirmation)	Ctrl+K
Open Configuration dialog	Ctrl+,
Open Contacts dialog	Ctrl+B
Send quick message preset 1–9	Alt+1 ... Alt+9

Main-window mnemonics

Widget	Mnemonic
GMRS radio	Alt+G
FRS radio	Alt+F
Clear chat button	Alt+C
Listen button	Alt+L
Transmit button	Alt+T
This is button	Alt+I
Dismiss all (pending pills)	Alt+D
Settings menu	Alt+S

Settings menu mnemonics

Item	Mnemonic
Configuration...	Alt+S, Alt+C
Contacts...	Alt+S, Alt+N
Quick Messages...	Alt+S, Alt+Q
Clear chat	Alt+S, Alt+R

Configuration dialog mnemonics

Field	Mnemonic
Callsign	Alt+C
Name	Alt+N
Location	Alt+L
Voice Model	Alt+V
Test voice	Alt+T
Speech Rate	Alt+R
Input Device	Alt+I
Output Device	Alt+O
VAD Threshold	Alt+D
Time Format	Alt+F
Filter profanity	Alt+Y
PTT Mode	Alt+P
Serial Port	Alt+S
Control Line	Alt+E

Tab order in the main window

Listen → Target dropdown → Message box → Transmit → This is. The order is explicitly set so keyboard-only operators get a predictable traversal.

11. GMRS vs FRS service mode

The top-of-window toggle controls which Part 95 subpart the app treats as canonical. GMRS (Subpart A) is licensed and callsign-bearing; FRS (Subpart B) is unlicensed and has no callsign requirement. Switching modes is instantaneous; saved contacts and the saved callsign are preserved across mode changes so switching back to GMRS restores prior state.

Feature	GMRS (default)	FRS
Header line	Station: CALL Operator Location	FRS Mode Operator Location
Outgoing TX preface	[CALL] [name] calling [target]	No preface — message is spoken as-is
15-minute ID rule	Enforced	Skipped
This is button	Enabled	Disabled (tooltip explains)
Target dropdown	Visible	Hidden
Contacts menu / icon	Enabled	Disabled (tooltip explains)
Pending-station pills	Detected and displayed	Detection suppressed
Callsign chat highlighting	Amber pills, FCC ✓	Suppressed
Online / Offline indicator	Visible (gates verification)	Hidden (no online features apply)
FCC callsign verification	On (opt-in via connectivity)	Not applicable

Why FRS turns features off: FRS operators are anonymous by regulation. Surfacing the 15-minute ID button or callsign verification on an FRS channel would be misleading at best and FCC-incorrect at worst. The toggle is therefore a hard behavioral switch, not a cosmetic relabel.

12. PTT (Push-to-Talk) modes

Selectable from Configuration → PTT Mode. The choice determines how the radio is keyed around each transmission.

Manual

Default. You press PTT on the radio yourself, the app plays audio through the configured output device, and you release PTT when the audio finishes. The app makes no attempt to key the radio.

Use when: you have a desk mic with a foot switch, you're operating a handheld and physically pressing the PTT button, or you're testing the app without any radio connected.

VOX (Voice-Operated Transmit)

Your radio is set to auto-key whenever it hears audio on its mic input. The app appends a short tail of silence after each transmission so the last syllable survives the VOX hang dropout.

Use when: your radio has a VOX setting and you've already tuned VOX sensitivity / hang time to your setup. Simplest wiring — just a cable from the output device to the radio's mic input.

USB FTDI / Serial

The app keys PTT through a USB-serial adapter's RTS or DTR line. The line drives an external transistor or opto-isolator wired to the radio's PTT pin. Short lead-in and tail silence are inserted around each transmission so the radio's keying ramp doesn't clip the audio.

- **Serial Port** — Device path — e.g. `/dev/ttyUSB0` on Linux or COM3 on Windows. Field is enabled only when this PTT mode is selected.
- **Control Line** — RTS or DTR. Choose whichever your interface uses. Most FTDI-based DIY PTT cables use RTS; Signalink-style boxes vary.

Cable safety: never wire a serial line directly to a radio's PTT pin. Use an opto-isolator or NPN transistor between the adapter and the radio so a stuck line or a host crash can't dump TTL voltage onto the radio.

13. Receive (Rx) pipeline

Pressing Listen starts a background thread that captures audio, gates it through Silero VAD, conditions each utterance through a DSP chain, and hands the result to faster-whisper for transcription. The pipeline is designed to skip kerchunks, static, and Whisper hallucinations so the chat log shows only genuine speech.

Audio capture

- On Linux with PipeWire the app prefers `parec` over PortAudio because PortAudio's PipeWire-via-ALSA bridge can silently deliver flat-zero audio on PipeWire 1.4. If `parec` is missing the app falls back to PortAudio.
- Anywhere else (or if you've selected a specific input device index in Configuration), PortAudio is used directly.

Voice activity detection (VAD)

- **Silero VAD** — Neural VAD. Only audio it scores as speech is forwarded for transcription — kerchunks and static are dropped.
- **Threshold** — Tunable from 0.10 (very sensitive) to 0.95 (very strict) in Configuration. Default 0.50.
- **Auto-rebaseline** — After roughly 30 s of continuous silence the VAD is rebaselined so detection stays responsive on long-quiet channels.
- **TX auto-pause** — While the app is transmitting, the listener is paused so your own TTS isn't transcribed back. Listening resumes immediately on unkey, with VAD state reset so no in-progress speech bleeds across the boundary.

DSP conditioning

- **300–3000 Hz bandpass** — Matches the narrowband-FM voice band. Strips hum and out-of-band hiss before denoising.
- **Noise reduction** — Spectral gating applied per utterance after bandpass and before transcription.
- **Hallucination filter** — Drops utterances shorter than ~400 ms and common Whisper hallucinations on silence (“Thank you”, “Subtitles by...”, etc.).

Transcription

- **Engine** — faster-whisper (CTranslate2 backend). int8 CPU inference by default.
- **Default model** — `small.en`. Trade quality / speed via `bootstrap_models.py --model base.en|medium.en`.
- **Profanity masking** — When **Filter profanity** is on, strong language in the transcript is masked with asterisks before it lands in the chat log.

14. Transmit (Tx) pipeline

Each transmission (typed message, quick preset, or This is) passes through the same pipeline.

- **1. Placeholder substitution** — Curly-brace tokens in quick presets are filled by prompting the user.
- **2. Shorthand expansion** — TTY and ham-radio shorthand is rewritten to full words before TTS — GA → Go ahead, SKSK → end of conversation, 73 → best regards, QTH → location, etc. Matching is case-insensitive and word-bounded, longest-key-first, so QSO → radio contact while a lone Q inside another word passes through.
- **3. Profanity masking** — When the filter is on (default), strong language is asterisk-masked before TTS sees it. Mild PG-13 words (damn, hell, crap, bare ass) pass through unchanged. Word-bounded so substrings like classroom or Scunthorpe are never false-positives.
- **4. FCC framing (GMRS only)** — Prepends [CALL] [name] calling [target] when targeting a specific station; appends a station ID if more than 15 minutes have passed since last identification. *All*-targeted transmissions skip the preface but still pick up the ID append when the rule triggers. FRS skips both.
- **5. Spoken-callsign formatting** — Digits inside callsigns are split with spaces so TTS reads them as individual digits (WSLZ 2 3 3 rather than two hundred thirty-three).
- **6. PTT key-down** — Manual mode: no-op. VOX: append silence tail. USB FTDI: assert the configured control line and pad with lead-in / tail silence.
- **7. STT auto-pause** — The listener is paused so your own audio isn't transcribed back.
- **8. Piper synthesis** — TTS audio is rendered offline through the selected Piper voice at the configured length scale and played to the configured output device.
- **9. Unkey and resume** — PTT releases (in serial mode) after a tail of silence; STT resumes with VAD state reset.

15. Callsign detection & FCC verification

Recognized formats

- GMRS modern: WSLZ233.
- GMRS legacy: KAE1234.
- US amateur: K1ABC, KD9XYZ, W1AW.
- Compact: WSLZ233.
- Spaced: W S L Z 2 3 3.
- Separated: W.S.L.Z.233, WSLZ-233, WSLZ, 233.
- NATO phonetic: Whiskey Sierra Lima Zulu Two Three Three (also accepts *X-ray* and *X ray*).

Pending pills

Unknown stations earn a yellow pill on the pending bar. The “unknown” check considers all three callsign fields on every contact (`callsign`, `gmrs_callsign`, `ham_callsign`), so a HAM call detected over the air won't pill if the operator's GMRS call is already saved (and vice versa).

FCC verification (online, opt-in)

When the connectivity probe says the app is online, contacts are cross-referenced against the public FCC license database via the `ke8rxnwx` crossref API. A row earns a green ✓ in the Verified column when both conditions hold:

- The callsign is in the active FCC database, AND
- The contact's name matches a token in the licensee's name (“Tim” matches “Surname, Timothy L”).

Verified lookups also persist `gmrs_callsign` and `ham_callsign` fields on the contact, pulled from the FCC related cross-reference list (service codes ZA for GMRS and HA / HV for Amateur). For an operator licensed for both services, a HAM call entered as the primary will resolve its associated GMRS call and vice versa.

Cross-references are **only** written when the contact's name matches the licensee. A family-member row on a shared GMRS call (whose name doesn't match) keeps its own GMRS / HAM fields untouched because those callsigns describe the licensee, not the family member.

A verified lookup also backfills the contact's `location` field with the FCC city (title-cased) when the row had no location of its own. Any value the operator already typed is left alone.

Caching: rows whose `verified` flag is `True` with unchanged callsign and name are treated as cached. Both `Verify all` and `OK` skip them, so opening and closing `Contacts` doesn't churn the `verified_at` timestamp. Newly added rows, in-dialog edits, and previously-failed lookups all earn a fresh round trip.

16. FCC Part 95 compliance features

GMRS-TTY makes Part 95 GMRS compliance the default behavior:

- Outbound messages always carry your callsign and name when targeting a specific station.
- The 15-minute ID rule is enforced automatically — your callsign and name are appended when more than 15 minutes have passed since the last identification.
- Identification is appended even on short messages if the rule triggers.
- The standalone **This is** button (Alt+I or Ctrl+I) sends a complete station ID and resets the 15-minute timer.
- The PG-13 profanity filter (default on) masks strong language in both RX and TX so transmissions stay within Part 95 obscenity expectations. Toggle in Configuration if you operate on a private repeater with different norms.
- FRS mode (Subpart B) intentionally skips Part 95 Subpart A station-ID rules. FRS is unlicensed and has no callsign requirement, so callsign framing, the 15-minute timer, and the standalone-ID button are all disabled.

You are still responsible for legal operation. This software does not replace a valid FCC GMRS license, and the operator remains accountable for content, channel choice, and any local repeater rules. The app's compliance features are a convenience, not a guarantee.

17. Accessibility (WCAG 2.1 AA)

The application exists for users with disabilities, so accessibility is a hard design constraint. The UI targets WCAG 2.1 Level AA — the practical baseline DOJ and Section 508 reference for software ADA compliance.

- **Color contrast** — Text colors meet $\geq 4.5:1$ against the chat background (RX green #15803D, TX blue #1D4ED8, errors #B91C1C, warnings #92400E). UI borders such as the pending-station pills meet $\geq 3:1$.
- **State never by color alone** — Every RX line is prefixed [RX HH:MM:SS] :, TX lines [TX to ...] : or [TX ID] :, errors say *Error:* or *Failed:* in the text. Color is supplemental, never the only cue.
- **Full keyboard operation** — Explicit tab order (Listen → Target → Message → Transmit → This is). Mnemonics on every actionable label. Global shortcuts listed in section 10.
- **Screen reader support** — Every non-decorative widget has an `accessibleName` and (where useful) `accessibleDescription` for the Qt accessibility bridge — NVDA, JAWS, Orca, VoiceOver.
- **Font scaling** — No hard-coded font-size in stylesheets. Bold uses Qt's relative QFont API so the OS font-scale setting carries through.
- **Resizable layout** — Main window minimum 720×520 so high-DPI / large-font setups don't clip. All dialogs are resizable.
- **Focus visible** — Relies on the Fusion style's default focus indicator — Qt does not strip outlines and neither does the app.

18. Off-grid operation

Once the dependencies and models are in place, the radio workflow does not need a network. RX transcription, TX synthesis, PTT keying, and contact management all run locally.

- **Whisper STT** — Loaded from `Models/STT/<model>/`. Pre-staged via `bootstrap_models.py`; never re-fetched at runtime.
- **Silero VAD** — Ships as a local ONNX file inside the wheel.
- **Piper TTS** — Voices live under `Voices/`; you supply them once.
- **Online features** — Strictly opt-in via the connectivity probe. When offline: the Verify all button disables, save-time verification is skipped, the online indicator turns amber, and previously-earned verified checks are preserved untouched. The radio workflow keeps running.

Recommended deployment for a true air-gapped target: clone the repo, install dependencies, and run `bootstrap_models.py` on an internet-connected machine. Then copy the entire source tree, `Models/`, and `Voices/` to the offline target. Reuse the same virtual environment if it's portable across the architecture, otherwise rebuild the venv from the same `requirements.txt` on the target.

19. Troubleshooting

Listen does nothing or fails immediately

- **“STT model not found” in chat** — Run `python bootstrap_models.py --model <name>` with the model named in the message.
- **Listen toggles but the level meter stays at zero** — The app isn't getting audio. Check the Input Device dropdown in Configuration, your microphone privacy permissions, and the physical cable. On PipeWire installs, confirm `pulseaudio-utils` is installed so `parec` is on PATH.
- **Lots of false starts on a noisy channel** — Raise VAD Threshold (try 0.65–0.75). Conversely, on a very quiet channel where speech is being missed, lower it (try 0.30).

Transmit is silent or clipped

- **Nothing audible at all** — Check Output Device. If you're feeding the radio directly, make sure you didn't accidentally leave Output Device on System Default — you'd hear it through the laptop speakers instead.
- **First syllable clipped on VOX** — VOX hang/sensitivity needs tuning on the radio; lengthen the radio's VOX attack if available.
- **Last syllable cut off on serial PTT** — Increase the tail-silence padding (recompile with longer pad) or switch to VOX where the tail is already extended.

FCC verification problems

- **Verify all is disabled** — Status bar shows Offline. Reconnect; the indicator refreshes every 30 s.
- **Row stays unverified even though the callsign is real** — Check that the Name field matches a token in the licensee's name. Hover the empty Verified cell — if the tooltip lists a licensee, the row was rejected for name-mismatch on purpose (family-shared GMRS call).

Quick messages

- **Strip is invisible** — You have no presets saved. Open the Quick Messages dialog and add at least one phrase.
- **Alt+1...Alt+9 fire the wrong preset** — Use Move Up / Move Down in the Quick Messages dialog to reorder — the shortcuts follow the saved order.

20. File reference

config.json

Lives in the working directory next to `main.py`. Plain UTF-8 JSON.

```
{
  "callsign": "WSLZ233",
  "name": "Benjamin",
  "location": "Lansing, MI",
  "voice": "Voices/en_US-ryan-high.onnx",
  "tts_length_scale": 1.0,
  "input_device": -1,
  "output_device": -1,
  "whisper_model": "small.en",
  "vad_threshold": 0.5,
  "time_format": "24h",
  "filter_profanity": true,
  "ptt_mode": "manual",
  "ptt_serial_port": "",
  "ptt_serial_line": "RTS",
  "radio_service": "GMRS",
  "quick_messages": [
    "Radio check",
    "Loud and clear",
    "Standing by",
    "Acknowledged",
    "Say again",
    "QSY to channel {N}",
    "Clear",
    "Monitoring",
    "Net check-in",
    "Emergency traffic"
  ]
}
```

- **input_device / output_device** — -1 means system default. Any other integer is the PortAudio device index.
- **tts_length_scale** — 0.70–1.50. Higher is slower.
- **vad_threshold** — 0.10–0.95. Higher is stricter.
- **ptt_mode** — `manual`, `vox`, or `usb_ftdi`.
- **ptt_serial_line** — `RTS` or `DTR`. Only used when `ptt_mode` is `usb_ftdi`.
- **time_format** — `24h` or `12h`.
- **radio_service** — `GMRS` or `FRS`. Default `GMRS` if missing or unknown.
- **quick_messages** — Ordered list of phrase strings; first nine get `Alt+1`...`Alt+9`.

contacts.json

Lives alongside `config.json`. Array of contact objects. The synthetic *All* row is always pinned at the top of the in-app dropdown and is preserved across saves.

```
[
  { "callsign": "All", "name": "Everyone" },
  {
    "callsign": "WSLZ233",
    "name": "Tim",
    "location": "Lansing, MI",
    "gmrs_callsign": "WSLZ233",
    "ham_callsign": "KD9XYZ",
    "verified": true,
    "verified_at": "2026-05-16T14:32:11Z",
    "license_name": "SURNAME, TIMOTHY L"
  }
]
```

- **callsign** — Required. Uppercased on save.
- **name** — Free-form.
- **location** — Free-form. Auto-backfilled from FCC city on successful verification if originally empty.
- **gmrs_callsign / ham_callsign** — Cross-references. Auto-populated by FCC verification when the name matches; hand-editable otherwise.
- **verified** — Boolean. True only when callsign is active in the FCC database AND name matches the licensee.
- **verified_at** — ISO-8601 UTC timestamp of the last successful lookup.
- **license_name** — Raw licensee name returned by the FCC — stored so the Verified-cell tooltip can show *FCC license is held by*

21. Glossary

Term	Definition
FCC Part 95	The section of US federal regulations covering the Personal Radio Services. Subpart A is GMRS; Subpart B is FRS.
FRS	Family Radio Service. Unlicensed, no callsign, lower power, fixed antennas only on most channels.
GMRS	General Mobile Radio Service. Licensed (one license covers an immediate family), callsign required, up to 50 W on repeater inputs.
kerchunk	A brief unmodulated key-up with no voice. Common when checking repeater access. Silero VAD is designed to ignore these.
NATO phonetic	The ICAO/NATO spelling alphabet (Alfa, Bravo, Charlie...). Used over voice radio to make letters unambiguous.
Piper	An offline neural TTS engine using ONNX voices.
PTT	Push-to-Talk. The control line that keys the radio into transmit mode.
QSL / QSO / QTH / QRZ	Ham-radio Q-signals. QSL = acknowledge. QSO = radio contact. QTH = location. QRZ = who is calling me?
Silero VAD	An open-source neural voice-activity detector. Decides which audio chunks contain speech.
STT	Speech-to-text. The Whisper transcription stage.
TTS	Text-to-speech. The Piper synthesis stage.
TTY	Teletypewriter — the device, and by extension the abbreviation style (GA, SK, 73, ILY), that deaf operators have used over phone and radio for decades.
VAD	Voice activity detection. See Silero VAD.
VOX	Voice-operated transmit. A radio's auto-key on detected audio.
Whisper	OpenAI's speech-recognition model. faster-whisper is the CTranslate2-accelerated CPU runtime used by this app.